Progress at a Glance

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Status on **23 October**: the architecture phase is complete. The project is moving into feature extraction and training. Every design decision is written down and the models are implemented.

Completed

- Literature check of all 10 proposal papers
- Full audit of the existing LSTM pipeline
- Dataset profiled: 874 ear videos, 643 / 231
- Data pipeline, LSTM and Transformer designs
- 7 model variants coded and verified (Keras 3.6)
- Benchmark and statistics protocol fixed

In progress / next

- MobileNetV2 feature cache (T = 40, once)
- Stratified 5-fold splits, saved and shared
- 90 runs: 6 models x 3 T values x 5 folds
- CPU benchmark harness (latency, RAM, FLOPs)
- Hypothesis tests and Pareto analysis
- Stretch: hybrid LSTM + Transformer

Research Questions + Null Hypothesis

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The question in testable form: given **identical frozen CNN frame features**, does a Transformer temporal head beat an LSTM head, and what does it cost on a **CPU**?

ID	Null hypothesis	How it is tested
H0-acc primary	Transformer AUROC = LSTM AUROC on short clips (T ≤ 40) with frozen MobileNetV2 features	Corrected resampled t-test on paired folds; TOST equivalence, margin 0.03 AUROC
H0-cpu	At matched parameters, Transformer batch-1 CPU latency ≤ LSTM latency	Median of ≥300 timed calls, 3 sessions, bootstrap CI on ratio
H0-T	Sequence length T (10, 20, 40) does not change AUROC for either family	Same paired tests across T, Holm correction

Why this null: Soeleman et al. 2022 (LSTM ≥ Transformer on 3 small video sets, same feature design); Barros et al. 2021 (CNN + LSTM, 185 ultrasound videos, ~93%); the Keras video Transformer example "works best with a larger dataset". Literature check: 6 of 10 papers verified, 2 partly, 2 need DOIs.

Dataset

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Measured property	Ear / otoscopy (primary)	Oral (optional)	External ear (optional)
Videos	**874**	535	50
Normal / Abnormal	**643 / 231** (26.4% abn.)	452 / 83 (15.5%)	29 / 21 (wax)
Majority-class accuracy	**73.6%**	84.5%	58.0%
Resolution	480 x 640 portrait, resized to 224 x 224		
Duration (median, range)	13.8 s (5.2 to 43 s)	25.4 s (1.3 to 59.7 s)	17.1 s (7 to 32 s)
Frames per video	146 to 1263	24 to 1280	204 to 946
Frame rate	14 to 30 fps	14 to 29 fps	19 to 30 fps
Age (years)	3 to 18, mostly 9 to 13	3 to 18	3 to 6
Size on disk	2.6 GB	3.2 GB	193 MB

- Ear is the single "type of videos" for the study; labels come from **labels.json** {id, interpretation, age}.
- **Abnormal = positive class**. Accuracy alone is misleading (73.6% for free), so **AUROC** is the primary metric.
- No patient ID in metadata: possible same-child leakage is documented and tested as a sensitivity check.

Audit of the Existing LSTM Pipeline

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Severity	Issue found in the old code	Effect	Fix in the new pipeline
HIGH	Pixels scaled /255 to [0,1]; MobileNetV2 expects [-1,1]	Backbone features off-distribution	mobilenet_v2.preprocess_input
HIGH	OpenCV BGR frames fed to an RGB model	Colour channels swapped	cvtColor BGR to RGB
HIGH	All frames held as float64: ~21 GB	Exceeds 16 GB RAM	uint8 frames, cached 1280-d features (179 MB)
HIGH	Confusion matrix run on every video, incl. training data	Inflated accuracy	Held-out folds only, out-of-fold predictions
MED	Plain KFold, accuracy only, Normal = 1	Misleading on 74 / 26 data	StratifiedKFold, AUROC, Abnormal = 1
MED	Videos with failed frame reads silently dropped	Biased, unlogged sample	Sequential decode, repeat-pad, logging
LOW	"MiniLSTM" is really a standard LSTM cell	Mislabelled baseline	Dropped; true minLSTM optional

Kept: the data, uniform 20-frame sampling, and the exact LSTM head (LSTM 256 to LSTM 128 to Dense 64). **Old accuracy numbers are not used as a baseline.**

Data Pipeline

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1. Video
.mp4, 480 x 640
~14 s, ~30 fps
labels.json

→

2. Sample frames
T = 40 uniform
sequential decode
BGR to RGB, 224^2
repeat-pad short
clips

→

3. MobileNetV2
ImageNet, frozen
preprocess_input
avg pool to 1280-d
~300 M MACs / frame

→

4. Feature cache
874 x 40 x 1280
float32, 179 MB
T=20: [::2]
T=10: [::4]

→

5. Splits
Stratified 5-fold
seeds 0, 1, 2
inner 15% val
shared JSON

Why cache the features?

Backbone frozen + no augmentation means identical outputs. MobileNetV2 runs **once** (~35k frames, ~10 to 20 min on CPU) instead of every epoch of every fold.

Fairness control

Every model reads the **same** feature file and the **same** split file. Only the temporal block differs, so differences come from that block alone.

System Architecture

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Input
frame features
(T, 1280)
from cache

→

Temporal block: the ONLY part that varies

LSTM LSTM(256, seq) → LSTM(128)
sequential over T, $O(T \cdot d^2)$

Transformer Dense(1280→d) + pos-emb → N x [MHA + FFN] → pool
parallel over T, $O(T \cdot d^2 + T^2 \cdot d)$

Hybrid (stretch) LSTM(128, seq) → 1 attention block → pool

→

Shared head
Dense(64, relu)
Dropout(0.5)
Dense(1, sigmoid)
P(Abnormal)

Held constant: features, folds, classifier head, loss (BCE + balanced class weights), epoch budget, early stopping on validation AUROC, seeds. **Varied:** temporal block and sequence length T.

LSTM Design

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$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i)$$
$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f)$$
$$g_t = \tanh(W_g x_t + U_g h_{t-1} + b_g)$$
$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o)$$
$$c_t = f_t \odot c_{t-1} + i_t \odot g_t$$
$$h_t = o_t \odot \tanh(c_t)$$

$$\text{params / layer} = 4(d_{in}h + h^2 + h)$$
$$\text{MACs / clip} = T \cdot 4(d_{in}h + h^2)$$

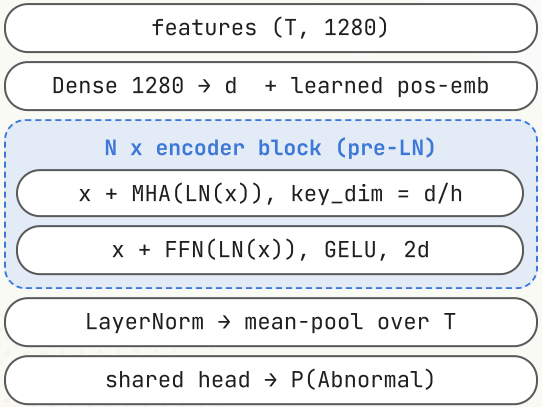
Preset	Layers	Params	MACs T=20
lstm_small	LSTM(128)	729,729	14.4 M
lstm_base	LSTM(256) → LSTM(128)	1,779,329	35.4 M
bilstm_base	BiLSTM(128) → BiLSTM(64)	1,615,489	32.1 M

- **lstm_base** = exact head of the existing model, now on correct features.
- 89% of lstm_base parameters sit in the first 1280 → 4x256 input kernel.
- T dependent steps per layer: poor multi-core use at batch 1.
- recurrent_dropout = 0 keeps the fused, fast LSTM kernel.

Recipe: Adam 1e-3, clipnorm 1.0, BCE, balanced class weights, batch 32, max 100 epochs, early stopping on val AUROC (patience 15), ReduceLROnPlateau.

Transformer Design

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Preset	d / heads / FFN / N	Params	MACs T=20
transformer_small	128 / 4 / 256 / 2	440,065	8.7 M
transformer_base	256 / 4 / 512 / 2	1,404,289	27.9 M
transformer_deep	256 / 8 / 512 / 4	2,458,497	49.3 M
hybrid_base	LSTM 128 + 1 block	862,465	17.1 M

Based on the Keras video Transformer example, with changes:

- Projection to d: in the example 4.2 M of 4.23 M params sit in attention.
- Pre-LN instead of post-LN: more stable on small data.
- Mean pooling instead of max; no CLS token to learn.
- **transformer_base** is parameter-matched to lstm_base (1.40 M vs 1.78 M).

Recipe: AdamW 3e-4, weight decay 1e-2, 5-epoch warm-up, dropout 0.1; same loss, batch, epoch budget and early stopping as the LSTM.

Compute Analysis (Pre-Experiment)

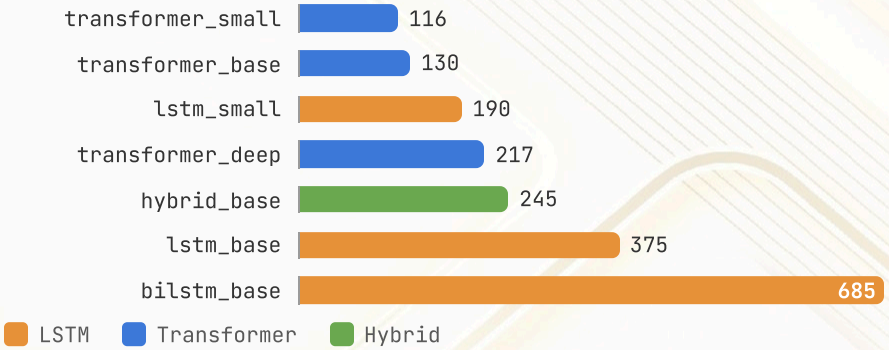
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MACs per clip	T=10	T=20	T=40	T^2 share
lstm_small	7.2 M	14.4 M	28.8 M	-
lstm_base	17.7 M	35.4 M	70.8 M	-
bilstm_base	16.1 M	32.1 M	64.2 M	-
transformer_small	4.3 M	8.7 M	17.9 M	4.6%
transformer_base	13.9 M	27.9 M	56.7 M	2.9%
transformer_deep	24.5 M	49.3 M	100.3 M	3.3%
MobileNetV2 backbone	3.0 G	6.0 G	12.0 G	-

- Attention T^2 term only dominates when T > 2d + FFN (512 frames at d=128).
- Temporal head is **< 1%** of end-to-end compute.

Smoke test: batch-1 latency, T = 20 (ms)

eager mode, untuned, single run. A hint, not a result.



On CPU at short T, the sequential LSTM loop may be **slower**, against the common belief. Testing this is H0-cpu.

Benchmarking + Evaluation Protocol

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Accuracy

- AUROC (primary), macro-F1, balanced accuracy
- Sensitivity (Abnormal recall), specificity, AUPRC
- Stratified 5-fold x up to 3 seeds
- Metrics only on held-out folds; out-of-fold predictions saved per video
- Grid: 6 models x T {10, 20, 40} x 5 folds = 90 runs

Compute (CPU only)

- Params; MACs (analytic + TF profiler)
- Latency: 50 warm-up + ≥300 timed calls, median / p95
- Batch 1 and 32; 1 and 8 threads; fresh process per model
- Peak RAM (RSS), train s / epoch, model file size
- Head-only and end-to-end with MobileNetV2

Machine: Intel Core i5-12450H (8 cores / 12 threads), 16 GB RAM, Windows 11, Python 3.11, TF 2.17, Keras 3.6
Statistics: corrected resampled t-test, 95% CI + Cohen's d, TOST (0.03 AUROC), McNemar + DeLong, Holm correction

Project Timeline + Status

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TODAY, OCT 23

Oct 5 — Architecture Finalisation

DONE

Designs for data, LSTM, Transformer and benchmark. 7 models coded.

Oct 12 — Data Preprocessing

DONE (DESIGN)

Dataset profiled and audited; fixed preprocessing specified.

Oct 19 — Training

IN PROGRESS

Feature cache, splits and k-fold loop; 90 runs on cached features.

Oct 26 — Benchmarking

UPCOMING

Latency, RAM, FLOPs, train time on CPU; 3 timing sessions.

Nov 2 — Final Analysis

UPCOMING

Hypothesis tests, Pareto plots, final presentation.

Nov 9 — Paper^

BUFFER

^If time permits: hybrid model + paper draft.

About one week behind the original plan. Caching features turns each training run from hours into minutes, which is how the schedule is recovered.

Codebase + Next Steps

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```
lstm-transformer-tradeoff/
├── configs/      data, lstm, transformer, benchmark
├── docs/         summary, abstract, scope, motivation
│   ├── design/   pipeline, LSTM, Transformer, protocol
│   └── research/ legacy audit, literature review
├── src/
│   ├── models/   heads, lstm, transformer, hybrid  [done]
│   ├── data/     video_io, features, splits        [next]
│   ├── training/ k-fold training loop              [next]
│   └── benchmark/ latency, memory, flops, report   [todo]
├── scripts/      model_summary                     [done]
├── data/         raw videos + feature cache (not in git)
└── results/      metrics, figures
```

Next step	Target
Frame sampler + MobileNetV2 feature cache	Oct 24
Saved stratified splits (3 seeds)	Oct 25
Training loop; first lstm_base fold	Oct 25
Full 90-run training grid	Oct 26 to 29
Benchmark harness + compute grid	Oct 29 to Nov 1
Statistics, Pareto plots, slides	Nov 2 to 6

Open items: DOIs for 2 proposal papers; confirm ear as the single dataset; any way to recover patient grouping.